

MACHINE LEARNING BASED BUYER SEGMENTATION AND INVESTMENT PROFILING FOR REAL ESTATE MARKET INTELLIGENCE

Research Paper

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Project Context: Unified Mentor × Parcel Co. Limited
Domain: Financial Analytics & Real Estate Market Intelligence

Machine Learning • K-Means • Hierarchical Clustering • Streamlit

Abstract

This research presents a machine-learning workflow for buyer segmentation and investment profiling in the real estate market. The study integrates client-level demographic, geographic, acquisition, financing and satisfaction information with property transaction data. The analytical dataset contains 2,000 buyers and 7,305 sold property records linked to clients. Data preparation includes duplicate checks, categorical normalization, robust date processing, age derivation, sale-price conversion and buyer-level aggregation of property purchasing behavior. Categorical variables are one-hot encoded and numerical variables are standardized before clustering. K-Means and hierarchical clustering are used to identify behavioral groups, while inertia and silhouette score are used to select the number of K-Means clusters. The evaluation selected three clusters, with the highest silhouette score among $k=2..8$. The resulting segmentation is exposed through a Streamlit dashboard containing buyer distribution, investor behavior, geographic analysis and segment-level insights. The study demonstrates how combining customer attributes with observed property-purchase behavior can support more differentiated real estate marketing and investment intelligence.

Keywords: Real Estate Analytics, Buyer Segmentation, K-Means, Hierarchical Clustering, Investment Profiling, Machine Learning, Streamlit

1. Introduction

Real estate markets contain heterogeneous buyers whose needs, financing patterns, geographic profiles and purchase motivations can differ substantially. Treating all buyers as one population can lead to generic recommendations, inefficient marketing allocation and limited understanding of investor behavior. The supplied Project Requirements Document (PRD) identifies the need for data-driven understanding of buyer types, investment motivations, geographic differences and financing patterns.

The project therefore frames buyer segmentation as an unsupervised learning problem. Rather than relying only on predefined labels, clustering is used to discover groups with similar behavioral and demographic characteristics. The workflow follows the PRD requirements for cleaning, feature encoding, scaling, K-Means clustering, hierarchical clustering, cluster-count evaluation and interpretation.

2. Problem Statement

The business problem is the lack of a unified analytical view of real estate buyers. The PRD highlights challenges including understanding different buyer types, investment motivations across demographics, geographic differences in investment behavior and customer financing patterns. These gaps can contribute to inefficient marketing spend, generic property recommendations and weak investor targeting.

3. Objectives

- Create a clean, integrated buyer-property analytical dataset.
- Engineer buyer-level indicators from property purchase history.
- Encode categorical variables and scale numerical variables for clustering.
- Evaluate K-Means solutions across multiple cluster counts using inertia and silhouette score.
- Use hierarchical clustering as a complementary segmentation approach.
- Profile the resulting buyer groups using investment, financing, demographic and satisfaction indicators.
- Provide an interactive Streamlit dashboard for live filtering and analytical exploration.

4. Dataset and Data Preparation

The project uses two related datasets: clients.csv and properties.csv. The client table contains customer identifiers, client type, gender, country, region, date of birth, acquisition purpose, loan application, referral channel and satisfaction score. The property table contains listing identifier, tower number, transaction date, unit category, unit number, floor area, sale price, listing status and client reference.

Dataset	Rows	Columns	Purpose
clients.csv	2,000	12	Buyer attributes and satisfaction
properties.csv	10,000	9	Property/listing and transaction attributes

Quality checks found no duplicate client rows and no duplicate property/listing rows. The property table contains 2,695 records without a client reference. The remaining 7,305 non-null references map to the client population used for buyer-level aggregation. Property sale prices are stored as currency text and are converted to numeric values during processing. Date values are parsed using a robust mixed-format approach.

The client population contains 1,897 Individual clients and 103 Company clients. Acquisition purpose is distributed between 1,385 Home and 615 Investment records. Loan application is recorded as Yes for 736 buyers and No for 1,264 buyers. The average satisfaction score is 3.03 on the five-point scale represented in the dataset.

5. Methodology

5.1 Data Cleaning

Cleaning includes duplicate removal, categorical whitespace normalization, robust date parsing, numeric conversion of currency-formatted sale prices and controlled handling of missing buyer-property relationships. The property aggregation is based on sold records, while buyers without a linked purchase are retained in the analytical buyer table with zero-valued purchase aggregates.

5.2 Feature Engineering

Age is derived from date of birth. Sold property records are aggregated at buyer level to create properties_purchased, total_investment, average_property_price, total_floor_area_sqft, average_floor_area_sqft, first_purchase_date, last_purchase_date, purchase_span_days and has_property_purchase. These features extend the PRD's customer-level variables with observed purchasing behavior.

Features used in the final clustering pipeline:

age, satisfaction_score, properties_purchased, total_investment, average_property_price, total_floor_area_sqft, average_floor_area_sqft, purchase_span_days, has_property_purchase, client_type, gender, country, region, acquisition_purpose, loan_applied, referral_channel

5.3 Encoding and Scaling

Categorical variables are transformed using one-hot encoding with unknown-category handling. Numerical variables are standardized using StandardScaler so that variables with large monetary or area magnitudes do not dominate distance-based clustering solely because of their scale.

5.4 Clustering

K-Means clustering is used as the primary segmentation model because it provides compact, interpretable partitions and is efficient for this dataset size. Hierarchical Agglomerative Clustering is also fitted as a complementary method to examine the consistency of the segmentation structure. The K-Means implementation uses a fixed random state and multiple initializations.

5.5 Cluster Selection

K-Means models are evaluated for $k=2$ through $k=8$. Inertia measures within-cluster compactness, while the silhouette score measures how well observations fit their own cluster relative to neighboring clusters. The PRD specifies the Elbow Method and Silhouette Score for optimal cluster selection.

6. Results

The evaluation selected $k=3$. Its silhouette score was 0.1670, which was higher than the scores obtained for the other tested values of k . The corresponding K-Means inertia was 16656.92.

k	Inertia	Silhouette Score
2	18,432.41	0.1658
3	16,656.92	0.1670
4	15,601.21	0.1077
5	14,832.25	0.0958
6	14,254.40	0.0912
7	13,775.16	0.0884
8	13,356.25	0.0877

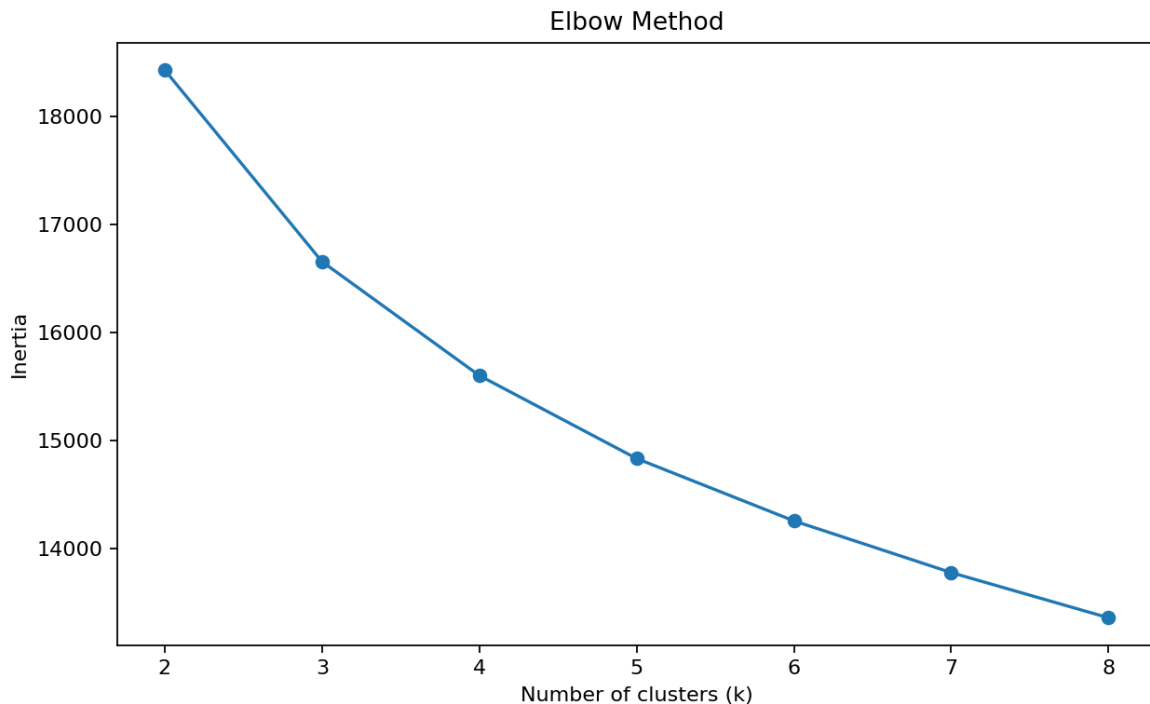


Figure 1. Elbow-method evaluation across candidate cluster counts.

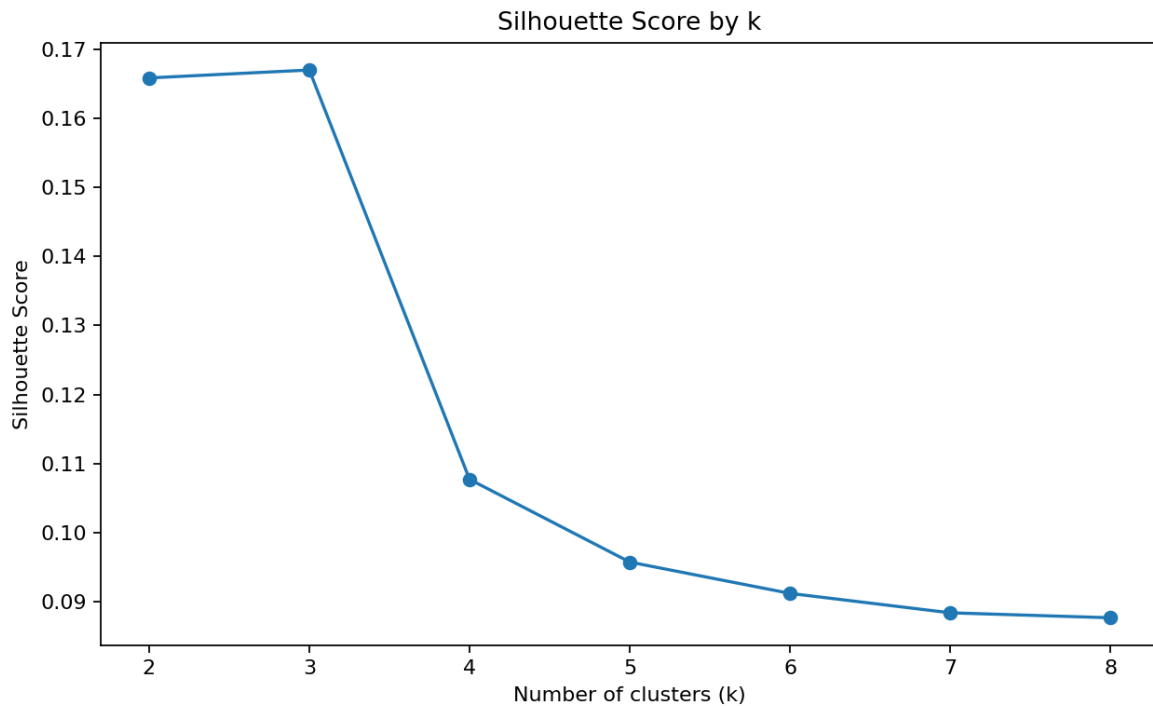


Figure 2. Silhouette scores across candidate cluster counts.

The final K-Means solution contains three clusters. The largest group contains 1,047 buyers, followed by 904 buyers, while the third group contains 49 buyers. This produces a strongly uneven segmentation, indicating that two broad groups cover most of the customer base while a smaller group exhibits distinct purchasing characteristics.

Cluster	Buyers	Avg Age	Avg Satisfaction	Avg Properties	Avg Total Investment	Avg Property Price	Loan Users	Investment Purpose %
0	1047	55.53	2.95	3.49	\$1,032,423	\$297,932	391	32.76%
1	904	55.09	3.10	3.65	\$1,460,716	\$404,712	329	28.21%
2	49	63.27	3.49	7.35	\$2,435,031	\$334,390	16	34.69%

6.1 Cluster 0

Cluster 0 contains 1,047 buyers. Its average age is 55.53, average satisfaction is 2.95, and average properties purchased is 3.49. Average total investment is \$1,032,423. Loan users account for 37.34% of the cluster and 32.76% of buyers have investment as the recorded acquisition purpose. This group represents a large mainstream buyer population with meaningful property-purchase activity.

6.2 Cluster 1

Cluster 1 contains 904 buyers and has an average age of 55.09. Average satisfaction is 3.10, average properties purchased is 3.65, and average total investment is \$1,460,716. The average property price is \$404,712. Loan users represent 36.39% of the cluster. Compared with Cluster 0, this group shows higher average investment and average property price.

6.3 Cluster 2

Cluster 2 is a small but distinct group of 49 buyers. It has the highest average age (63.27), highest average satisfaction (3.49), highest average properties purchased (7.35) and highest average total investment

(\$2,435,031). Its average property price is \$334,390. Only 32.65% are loan users. These observed characteristics indicate a relatively high-value, repeat-purchase segment.

7. Exploratory Data Analysis

Exploratory analysis was used to understand the composition of the customer base before clustering. The figures below are generated directly from the project pipeline.

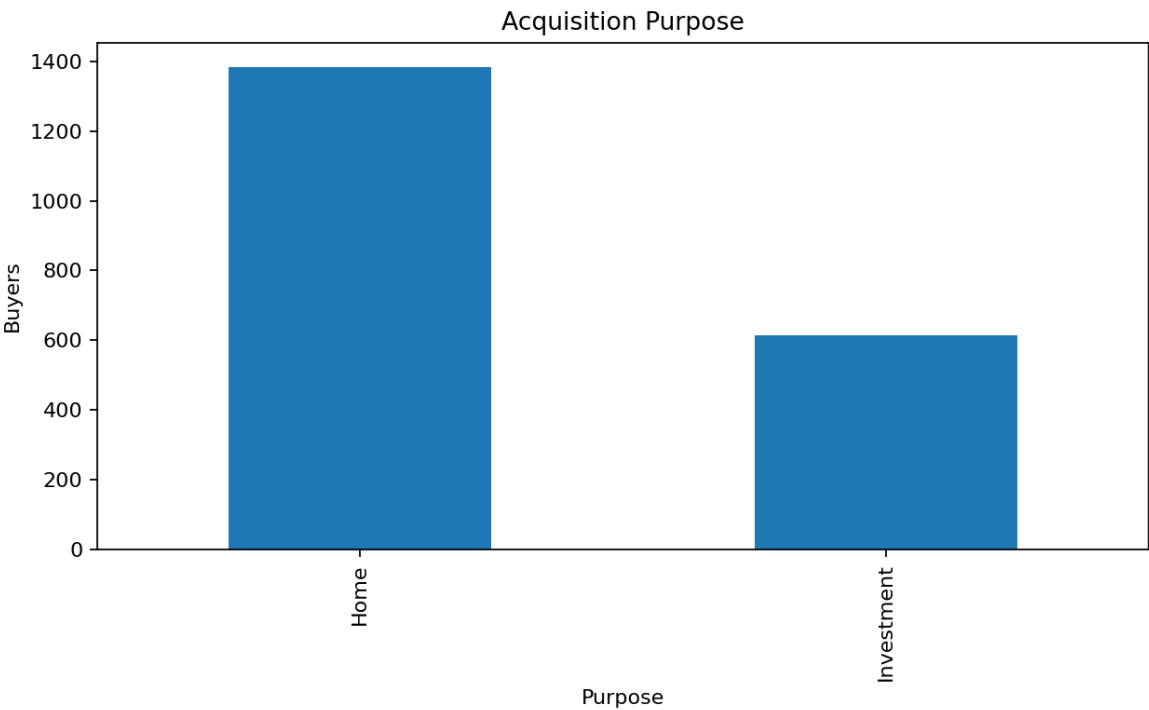


Figure 3. Distribution of buyers by acquisition purpose.

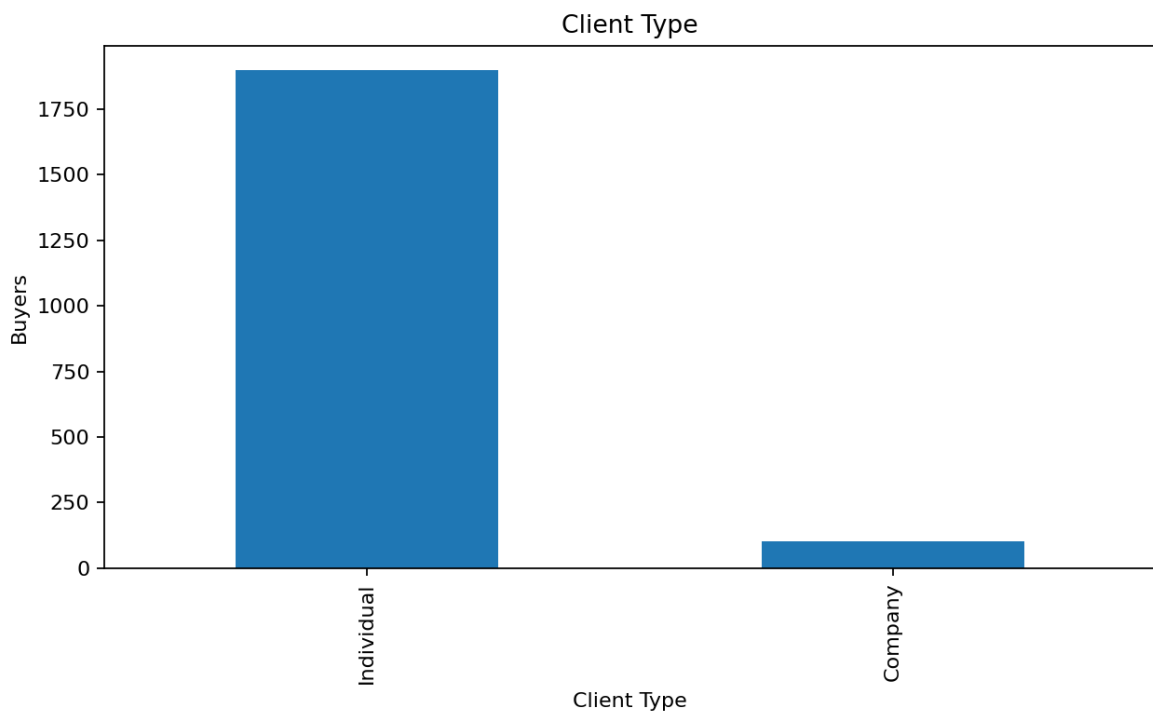


Figure 4. Distribution of buyers by client type.

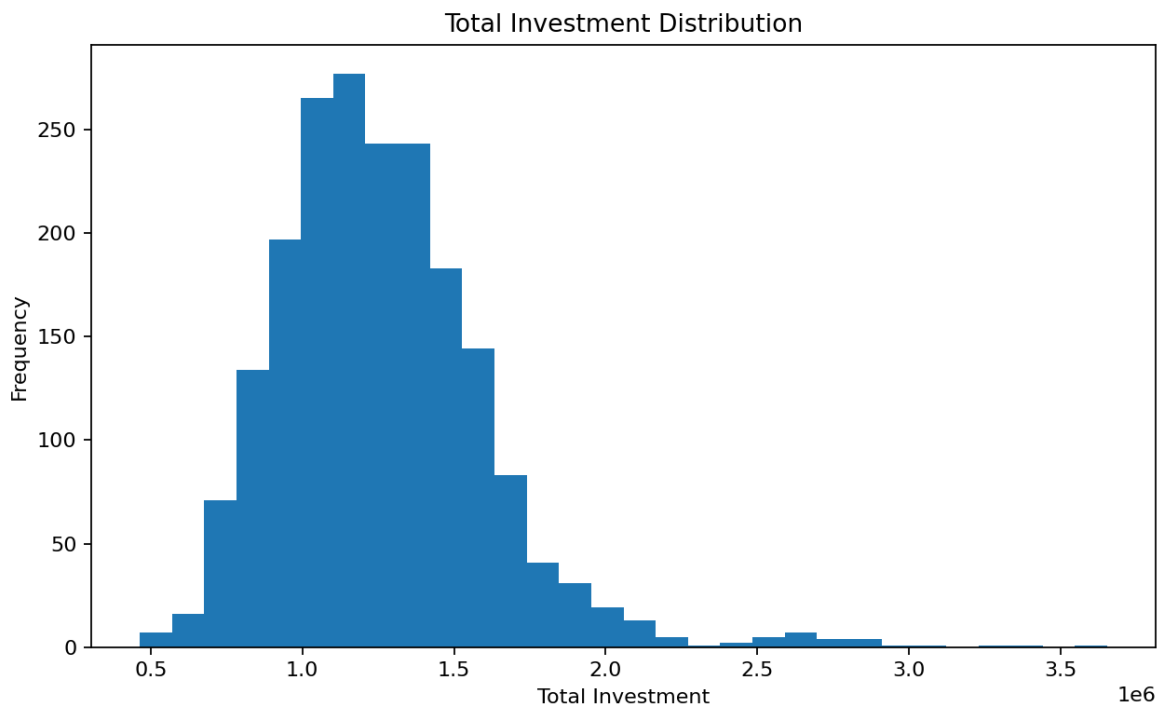


Figure 5. Distribution of buyer-level total investment.

8. Business Insights

- Buyer behavior is heterogeneous enough to justify segmentation rather than one-size-fits-all targeting.

- The largest two clusters account for most buyers, while a small third cluster shows substantially higher purchase frequency and investment value.
- Cluster 2 combines higher purchase frequency, higher average investment and lower loan usage, making it particularly relevant for high-value investor targeting.
- Investment purpose and loan behavior provide useful dimensions for interpreting buyer profiles, but they should be combined with observed property-purchase behavior rather than used alone.
- Geographic and referral-channel variables can support targeted campaigns because the analytical dataset preserves country, region and acquisition-channel information.

9. Recommendations

Recommendation	Application
Segment-specific marketing	Use the cluster profiles to create differentiated campaign messages instead of sending identical offers to all buyers.
High-value retention	Prioritize Cluster 2 for relationship management, premium listings, repeat-purchase programs and investor-focused communications.
Investment targeting	Use acquisition purpose, investment amount and purchase frequency together to identify customers with stronger investment orientation.
Financing strategy	Use loan behavior to separate financing-sensitive audiences from buyers who appear less dependent on loans.
Geographic personalization	Combine region and country with cluster membership to adapt property recommendations and campaign timing.
Dashboard-driven decisions	Use the Streamlit dashboard for interactive filtering and descriptive monitoring before making campaign decisions.

10. Streamlit Dashboard

The project includes a Streamlit dashboard designed as the operational analytics layer. It presents buyer segmentation overview, investor behavior, geographic buyer analysis and a segment insights panel. Users can filter the analytical data by country, region, acquisition purpose and client type. The dashboard also displays key performance indicators including the 2,000 buyer population, 7,305 linked sold purchases, total investment of \$2,520,750,961, and average satisfaction of 3.03.

A live Indian Standard Time clock and project ownership information are included in the interface. The dashboard is intended for exploratory and management use, while the underlying pipeline remains reproducible through `run_pipeline.py`.

11. Limitations

- The available dataset does not provide a direct income field, so income-related buyer classifications should not be claimed solely from the clustering output.
- The PRD describes recommended buyer types such as global investors, first-time buyers, corporate buyers and luxury investors, but the final unsupervised model contains three observed clusters; therefore those labels should not be assigned automatically without additional validation.
- The silhouette score is relatively low in absolute terms, so the three-cluster solution should be treated as a useful segmentation structure rather than a perfect separation of buyer populations.
- Geographic interpretation is descriptive unless spatial coordinates or additional market variables are introduced.

- Clustering results are dependent on feature selection, preprocessing and scaling choices.

12. Future Scope

- Add income, property valuation, rental yield, mortgage amount and portfolio value features when available.
- Evaluate additional clustering algorithms such as DBSCAN or Gaussian Mixture Models.
- Add dimensionality reduction such as PCA for visual cluster exploration.
- Introduce time-series analysis of buyer acquisition and purchasing behavior.
- Add geospatial maps and market-level indicators.
- Introduce model monitoring and periodic retraining as new buyer and property transactions arrive.
- Validate cluster stability using repeated runs, bootstrap analysis or external business labels.

13. Conclusion

This project demonstrates an end-to-end machine-learning approach to real estate buyer segmentation and investment profiling. The integration of client attributes with buyer-level property-purchase behavior provides a richer analytical representation than demographic information alone. After cleaning, feature engineering, one-hot encoding and standardization, K-Means and hierarchical clustering were applied. Evaluation across $k=2..8$ selected three K-Means clusters based on the highest observed silhouette score. The final segmentation reveals two large buyer groups and a smaller, higher-value repeat-purchase group. The Streamlit dashboard translates these analytical outputs into an interactive market-intelligence interface. Overall, the workflow provides a practical foundation for differentiated marketing, investor targeting and data-driven real estate decision-making.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Saiprasad Kagne, my project mentor, for valuable guidance, support, and constructive feedback throughout the development of this project, “Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence.”

References

- [1] Unified Mentor and Parcel Co. Limited, “Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence,” Project Requirements Document, 2026.
- [2] Project analytical datasets: clients.csv and properties.csv, supplied for the project implementation.
- [3] Project-generated artifacts: buyer_analytical_dataset.csv, buyer_clustered_dataset.csv, cluster_evaluation.csv and cluster_summary.csv.
- [4] Project implementation files: data_pipeline.py, model.py, eda.py, run_pipeline.py and app.py.

Appendix A. Reproducibility

From the project root, activate the virtual environment and run:

```
$env:MPLBACKEND="Agg"  
python run_pipeline.py  
python -m streamlit run app.py
```